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Electronic device controlling pulse signal from processor to display

Abstract

An electronic device is provided. The electronic device includes a processor. The electronic device includes a display including a display driver circuit and a display panel. The electronic device includes a first path connecting the display driver circuit to the processor. The electronic device includes a second path connecting the display driver circuit to the processor and separate from the first path. The processor is configured to transmit, based on a first cycle, via the second path to the display driver circuit, a pulse signal to synchronize, with a first time period in the processor used to display an image transmitted via the first path to the display driver circuit on the display panel, the first time period in the display driver circuit used to display the image on the display panel. The processor is configured to change, based on a second cycle different from the first cycle, a waveform of the pulse signal transmitted from the first cycle from a first waveform to a second waveform to synchronize, with a second time period in the processor used for the display of the image, the second time period in the display driver circuit used for the display of the image.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of PCT International Application No. PCT/KR2023/014940, which was filed on Sep. 26, 2023, and claims priority to Korean Patent Application No. 10-2022-0125365, filed on Sep. 30, 2022, in the Korean Intellectual Property Office and claims priority to Korean Patent Application No. 10-2023-0001471, filed on Jan. 4, 2023, in the Korean Intellectual Property Office and claims priority to Korean Patent Application No. 10-2023-0004350, filed on Jan. 11, 2023, in the Korean Intellectual Property Office and claims priority to Korean Patent Application No. 10-2023-0013290, filed on Jan. 31, 2023, in the Korean Intellectual Property Office and claims priority to Korean Patent Application No. 10-2023-0026723, filed on Feb. 28, 2023, in the Korean Intellectual Property Office and claims priority to PCT International Application No. PCT/KR2023/014711, filed on Sep. 25, 2023, in the Korean Intellectual Property Office, the disclosures of each of which are incorporated by reference herein their entireties.

BACKGROUND
Field
(1) The disclosure relates to an electronic device controlling a pulse signal from a processor to a display.
Description of Related Art
(2) An electronic device may include a display panel. For example, the electronic device may include a display driver circuit operably coupled with the display panel. For example, the display driver circuit may display an image obtained from a processor of the electronic device on the display panel.
(3) The above-described information may be provided as a related art for the purpose of helping to understand the present disclosure. No claim or determination is raised as to whether any of the above-described information can be applied as a prior art related to the present disclosure.
SUMMARY
(4) An electronic device is provided. The electronic device may comprise a processor. The electronic device may comprise a display including a display driver circuit and a display panel. The electronic device may comprise a first path connecting the display driver circuit to the processor.

The electronic device may comprise a second path connecting the display driver circuit to the processor and separate from the first path. The processor may be configured to transmit, based on a first cycle, via the second path to the display driver circuit, a pulse signal to synchronize, with a first time period in the processor used to display an image transmitted via the first path to the display driver circuit on the display panel, the first time period in the display driver circuit being used to display the image on the display panel. The processor may be configured to change, based on a second cycle different from the first cycle, a waveform of the pulse signal transmitted from the first cycle from a first waveform to a second waveform to synchronize, with a second time period in the processor used for the display of the image, the second time period in the display driver circuit being used for the display of the image.
(5) An electronic device is provided. The electronic device may comprise a processor. The electronic device may comprise a display including a display driver circuit and a display panel. The electronic device may comprise a first path connecting the display driver circuit to the processor. The electronic device may comprise a second path connecting the display driver circuit to the processor and separate from the first path. The processor may be configured to periodically transmit, via the second path to the display driver circuit, a pulse signal to synchronize, with a time period in the processor used to display on the display panel an image transmitted via the first path to the display driver circuit, the time period in the display driver circuit being used to display on the display panel the image. The processor may be configured to identify a control command to be provided to the display driver circuit with respect to the display of the image on the display panel. The processor may be configured to, based on the identification, change a waveform of the pulse signal that is periodically transmitted to the display driver circuit from a first waveform to a second waveform to indicate the control command to the display driver circuit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The above and other aspects, features and advantages of certain embodiments of the present disclosure will be more apparent from the following detailed description, taken in conjunction with the accompanying drawings, in which:
(2) FIG. **1** is a simplified block diagram of an exemplary electronic device.
(3) FIG. **2** illustrates an example of a first path and a second path each connecting a processor and a display driving circuit.
(4) FIG. **3** illustrates an example of a pulse signal indicating a timing of a horizontal synchronization signal and a timing of an emission synchronization signal.
(5) FIG. **4** illustrates an example of a pulse signal indicating a timing of a horizontal synchronization signal and a timing of a vertical synchronization signal.
(6) FIG. **5** illustrates an exemplary method of changing a waveform of a pulse signal according to a change in a timing of a vertical synchronization signal.
(7) FIG. **6** illustrates an exemplary method of changing a waveform of a pulse signal to indicate a control command.
(8) FIG. **7** is a block diagram of an electronic device in a network environment, according to various embodiments.
(9) FIG. **8** is a block diagram of a display module, according to various embodiments.
DETAILED DESCRIPTION
(10) FIG. **1** is a simplified block diagram of an exemplary electronic device.
(11) Referring to FIG. **1**, an electronic device **100** may include a processor (e.g., including processing circuitry) **120** and a display **130**. The display **130** may include a display driver circuit **131** and a display panel **132**.

(12) The processor **120** may include at least a portion of a processor **720** of FIG. **7**. The display **130** may include at least a portion of a display module **760**. The display driver circuit **131** may include at least a portion of a DDI **830** of FIG. **8**. The display panel **132** may include at least a portion of a display **810** of FIG. **8**.
(13) The processor **120** may be operably coupled with the display **130** (or the display driver circuit **131**). The operative coupling of the processor **120** and the display **130** (or the display driver circuit **131**) may indicate that the processor **120** is directly connected with the display **130** (or the display driver circuit **131**). The operative coupling of the processor **120** and the display **130** (or the display driver circuit **131**) may indicate that the processor **120** is connected with the display **130** (or the display driver circuit **131**) through another component of the electronic device **100**.
(14) The operative coupling of the processor **120** and the display **130** (or the display driver circuit **131**) may indicate that the processor **120** at least partially controls the display **130** (or the display driver circuit **131**) for a display of an image. The operative coupling of the processor **120** and the display **130** (or the display driver circuit **131**) may indicate that the processor **120** transmit an image acquired or rendered by the processor **120** to the display driver circuit **130** (or the display driver circuit **131**) in order for a display on the display panel **132**. However, it is not limited thereto.
(15) For example, the processor **120** is operably coupled with the display **130**, but a portion of operations of the processor **120** may not be synchronized with a portion of operations of display **130**.
(16) As a non-limiting example, the portion of the operations of the display **130** are unnoticeable or transparent to the processor **120**, the portion of the operations of the display **130** may not be synchronized with the portion of the operations of the processor **120**. As a non-limiting example, since the portion of the operations of the display **130** are independent of the processor **120**, the portion of the operations of the display **130** may not be synchronized with the portion of the operations of the processor **120**. As a non-limiting example, since the portion of the operations of the display **130** are performed or executed while the processor **120** is in a disabled state (e.g., a low-power state, a sleep state, and/or a turn-off state), the portion of the operations of the display **130** may not be synchronized with the portion of the operations of the processor **120**. As a non-limiting example, since the portion of the operations of the display **130** are performed or executed while an image transmission from the processor **120** to display **130** is ceased, the portion of the operations of the display **130** may not be synchronized with the portion of the operations of the processor **120**. As a non-limiting example, since the portion of the operations of the display **130** are performed or executed while a path used for the transmission is disabled, the portion of the operations of the display **130** may not be synchronized with the portion of the operations of the processor **120**. As a non-limiting example, the portion of the operations may include displaying an image stored in a graphic random access memory (GRAM) (e.g., a GRAM **220** of FIG. **2** or a memory **833** of FIG. **8**) in the display driver circuit **131** on the display panel **132**, while the transmission is ceased or the path is disabled.
(17) As a non-limiting example, that the portion of the operations of the processor **120** are not synchronized with the portion of the operations of the display **130** may be caused by a reference time of the display **130** (or the display driver circuit **131**) that is not synchronized with a reference time of the processor **120**. As a non-limiting example, that the portion of the operations of the processor **120** are not synchronized with the portion of the operations of the display **130** may be caused by an operation of a clock for the display **130** (or the display driver circuit **131**) that is not synchronized with an operation of a clock for the processor **120**. As a non-limiting example, that the portion of the operations of the processor **120** are not synchronized with the portion of the operations of the display **130** may be caused by counting of the display driver circuit **131** that is not synchronized with counting of the processor **120** related to a display on the display panel **132**.
(18) For example, the processor **120** may periodically transmit a pulse signal to the display driver circuit **131**, to synchronize the portion of the operations of the display **130** with the portion of the

operations of the processor **120**. As a non-limiting example, since the pulse signal is transmitted from outside (e.g., the processor **120**) of the display **130** for a synchronization of the display **130**, the pulse signal may be referred to as an external synchronization signal (Esync).
(19) For example, a transmission path of the pulse signal may be different from a transmission path of an image to be displayed on the display panel **132**. For example, the pulse signal may be transmitted from the processor **120** to the display driver circuit **131**, via a second path different from a first path used to transmit the image to be displayed on the display panel **132** from the processor **120** to the display driver circuit **131**. The first path and the second path may be illustrated in greater detail below with reference to FIG. **2**.
(20) FIG. **2** illustrates an example of a first path and a second path each connecting a processor and a display driving circuit.
(21) Referring to FIG. **2**, the electronic device **100** may include a first path **201** connecting the display driver circuit **131** with the processor **120** and a second path **202** connecting the display driver circuit **131** with the processor **120** and separated from the first path **201**.
(22) The first path **201** may be used to transmit an image to be displayed on the display panel **132** (not shown in FIG. **2**). For example, the processor **120** may transmit the image to be displayed on the display panel **132** to the display driver circuit **131** through the first path **201**. For example, the first path **201** may include a mobile industry process interface (MIPI). For example, the first path **201** may be enabled within a time period (or a time interval) in which the image to be displayed on the display panel **132** is transmitted from the processor **120** to the display driver circuit **131**, as indicated by a state **211**. For example, the first path **201** may be disabled within at least a portion of a time period (or a time interval) in which the transmission is ceased, as indicated by a state **212**. For example, the first path **201** may be disabled within at least a portion of a time period (or a time interval) in which the display driver circuit **131** executes a display on the display panel **132** based on a scan of an image stored in the GRAM **220** in the display driver circuit **131**, as indicated by a state **212**.
(23) As a non-limiting example, enablement of the first path **201** may indicate that normal power is provided to the first path **201**, and disablement of the first path **201** may indicate that low power is provided to the first path **201** or that providing power to the first path **201** is ceased. As a non-limiting example, the enablement of the first path **201** and the disablement of the first path **201** may be executed based on a control of the processor **120**.
(24) The second path **202** may be used for a transmission of the pulse signal. For example, transmitting the pulse signal through the second path **202** may be maintained even when transmitting an image through the first path **201** is ceased. For example, transmitting the pulse signal through the second path **202** may be maintained even when the first path **201** is disabled. For example, transmitting the pulse signal through the second path **202** may be maintained while an image stored in the GRAM **220** is scanned by the display driver circuit **131**.
(25) According to various embodiments, the pulse signal may be transmitted from the processor **120** to the display driver circuit **131** through the first path **201** while the first path **201** is enabled, and may be transmitted from the processor **110** to the display driver circuit **131** through the second path **202** while the first path **201** is disabled. For example, the processor **120** may adaptively identify a transmission path of the pulse signal from among the first path **201** and the second path **202**, according to a state of the first path **201**.
(26) According to an embodiment, the second path **202** may be a dedicated path for (only) the pulse signal. According to an embodiment, the second path **202** may be used for a signal transmitted from the display driver circuit **131** to the processor **120** as well as the pulse signal. For example, the signal may be a signal indicating a state of the display driver circuit **131**, which is transmitted from the display driver circuit **131** to the processor **120**. For example, the signal may be a tearing effect (TE) signal. However, it is not limited thereto.
(27) Referring back to FIG. **1**, a transmission cycle (or transmission interval) of the pulse signal

may correspond to a cycle (or interval) of a synchronization signal for the processor **120** used or identified for a display on the display panel **132**. As a non-limiting example, the transmission cycle may correspond to a cycle of a horizontal synchronization signal for the processor **120** used for the display on the display panel **132**. As a non-limiting example, the transmission cycle may correspond to a cycle of an emission synchronization signal for the processor **120** indicating a timing of an emission signal from the display driver circuit **131** to the display panel **132**.
(28) A waveform of the pulse signal may be changed to indicate a cycle of another synchronization signal for the processor **120**, wherein the other synchronization signal is distinct from the synchronization signal indicated by the transmission cycle and is used or identified for a display on the display panel **132**. As a non-limiting example, when the synchronization signal for the processor **120** indicated by the transmission cycle is the horizontal synchronization signal for the processor **120**, the waveform may be changed based on a vertical synchronization signal for the processor **120** and/or the cycle of the emission synchronization signal for the processor **120**. For example, the waveform of the pulse signal transmitted at a start timing of the horizontal synchronization signal for the processor **120** corresponding (or overlapping) to a start timing of the vertical synchronization signal (and/or the emission synchronization signal) for the processor **120** may be a second waveform (and/or a third waveform) different from the first waveform, which is the waveform of the pulse signal transmitted at the start timing of the horizontal synchronization signal for the processor **120** different from (or not overlapping) the start timing of the vertical synchronization signal for the processor **120** and/or the emission synchronization signal for the processor **120**. As a non-limiting example, when the synchronization signal for the processor **120** indicated by the transmission cycle is the emission synchronization signal for the processor **120**, the waveform may be changed based on the cycle of the vertical synchronization signal for the processor **120**. For example, the waveform of the pulse signal transmitted at the start timing of the emission synchronization signal for the processor **120** corresponding (or overlapping) to the start timing of the vertical synchronization signal for the processor **120** may be the second waveform different from the first waveform, which is the waveform of the pulse signal transmitted at the start timing of the emission synchronization signal for the processor **120** different from (or not overlapping) the start timing of the vertical synchronization signal for the processor **120**.
(29) The transmission cycle of the pulse signal and the waveform of the pulse signal may be illustrated in greater detail below with reference to FIGS. **3**, **4** and **5**.
(30) FIG. **3** illustrates an example of a pulse signal indicating a timing of a horizontal synchronization signal and a timing of an emission synchronization signal.
(31) Referring to FIG. **3**, a transmission cycle **301** of a pulse signal **393** may correspond to a cycle **302** of a horizontal synchronization signal **391** for the processor **120**. For example, the transmission cycle **301** of the pulse signal **393** may indicate the cycle **302** of the horizontal synchronization signal **391**. For example, the pulse signal **393** may have the transmission cycle **301**, to synchronize, with a first time period in the processor **120** corresponding to the cycle **302** of the horizontal synchronization signal **391**, the first time period in the display driver circuit **131** corresponding to a cycle of a horizontal synchronization signal for the display driver circuit **131**.
(32) For example, the waveform of the pulse signal **393** may be changed, based at least in portion on whether a start timing of the horizontal synchronization signal **391** overlaps (or corresponds) a start timing of an emission synchronization signal **392** for the processor **120**. For example, a width of the pulse signal **393** transmitted at the start timing of the horizontal synchronization signal **391** overlapping the start timing of the emission synchronization signal **392** may be different from a width of the pulse signal **393** transmitted at the start timing of the horizontal synchronization signal **391** that does not overlap with the start timing of the emission synchronization signal **392**.
(33) For example, the waveform of the pulse signal **393** transmitted from the processor **120** to the display driver circuit **131** at the start timing **321** of the horizontal synchronization signal **391** overlapping with the start timing of the emission synchronization signal **392** may be the second

waveform different first waveform, which is a waveform of the pulse signal **393** transmitted at the start timing **322** of the horizontal synchronization signal **391** that does not overlap with the start timing of the emission synchronization signal **392**. For example, a width **312** of the pulse signal **393** transmitted at the start timing **321** of the horizontal synchronization signal **391** may be different from a width **311** of the pulse signal **393** transmitted at the start timing **322** of the horizontal synchronization signal **391**. For example, the processor **120** may change the waveform of the pulse signal **393**, by changing the width of the pulse signal **393** to be transmitted at the start timing **321** of the horizontal synchronization signal **391** from the width **311** to the width **312**. For example, the processor **120** may change the waveform of the pulse signal **393**, by changing the width of the pulse signal **393** to be transmitted at the start timing **322** of the horizontal synchronization signal **391** from the width **312** to the width **311**.

(34) For example, since the start timing of the emission synchronization signal **392** identified according to counting of the horizontal synchronization signal **391** in the processor **120** and the start timing of the emission synchronization signal for the display driver circuit **131** identified according to counting of the horizontal synchronization signal for the display driver circuit **131** in the display driver circuit **131** may be changed according to a state of the processor **120** and/or a state of the display driver circuit **131**, the processor **120** may provide information on the emission synchronization signal **392** to the display driver circuit **131** through the change of the waveform.

(35) As a non-limiting example, when the processor **120** and the display **130** execute an image update according to the timing (or a transmission timing) of the emission signal, the pulse signal **393** may indicate the cycle **302** of the horizontal synchronization signal **391** through the transmission cycle **301**, and the cycle **303** of the emission synchronization signal **392** through a width (e.g., the width **311** or the width **312**). However, it is not limited thereto.

(36) FIG. **4** illustrates an example of a pulse signal indicating a timing of a horizontal synchronization signal and a timing of a vertical synchronization signal.

(37) Referring to FIG. **4**, a transmission cycle **401** of a pulse signal **493** may correspond to a cycle **402** of a horizontal synchronization signal **491** for the processor **120**. For example, the transmission cycle **401** of the pulse signal **493** may indicate the cycle **402** of the horizontal synchronization signal **491**. For example, the pulse signal **493** may have the transmission cycle **401**, to synchronize, with a first time period in the processor **120** corresponding to the cycle **402** of the horizontal synchronization signal **491**, the first time period in the display driver circuit **131** corresponding to a cycle of a horizontal synchronization signal for the display driver circuit **131**.

(38) For example, the waveform of the pulse signal **493** may be changed, based at least in portion on whether a start timing of the horizontal synchronization signal **491** overlaps (or corresponds) a start timing of a vertical synchronization signal **492** for the processor **120**. For example, a width of the pulse signal **493** transmitted at the start timing of the horizontal synchronization signal **491** overlapping the start timing of the vertical synchronization signal **492** may be different from a width of the pulse signal **493** transmitted at the start timing of the horizontal synchronization signal **491** that does not overlap with the start timing of the vertical synchronization signal **492**.

(39) For example, the waveform of the pulse signal **493** transmitted from the processor **120** to the display driver circuit **131** at the start timing **421** of the horizontal synchronization signal **491** overlapping with the start timing of the vertical synchronization signal **492** may be the second waveform different first waveform, which is a waveform of the pulse signal **493** transmitted at the start timing **422** of the horizontal synchronization signal **491** that does not overlap with the start timing of the vertical synchronization signal **492**. For example, a width **412** of the pulse signal **493** transmitted at the start timing **421** of the horizontal synchronization signal **491** may be different from a width **411** of the pulse signal **493** transmitted at the start timing **422** of the horizontal synchronization signal **491**. For example, the processor **120** may change the waveform of the pulse signal **493**, by changing the width of the pulse signal **493** to be transmitted at the start timing **421** of the horizontal synchronization signal **491** from the width **411** to the width **412**. For example, the

processor **120** may change the waveform of the pulse signal **493**, by changing the width of the pulse signal **493** to be transmitted at the start timing **422** of the horizontal synchronization signal **491** from the width **412** to the width **411**.
(40) For example, since the start timing of the vertical synchronization signal **492** identified according to counting of the horizontal synchronization signal **491** in the processor **120** and the start timing of the vertical synchronization signal for the display driver circuit **131** identified according to counting of the horizontal synchronization signal for the display driver circuit **131** in the display driver circuit **131** may be changed according to a state of the processor **120** and/or a state of the display driver circuit **131**, the processor **120** may provide information on the vertical synchronization signal **492** to the display driver circuit **131** through the change of the waveform.
(41) FIG. **5** illustrates an exemplary method of changing a waveform of a pulse signal according to a change in a timing of a vertical synchronization signal.
(42) Referring to FIG. **5**, the processor **120** may transmit a pulse signal **594** to the display driver circuit **131** based on a transmission cycle **501**, to indicate a cycle **502** (or a timing (or start timing) of a horizontal synchronization signal **591**) of a horizontal synchronization signal **591** for the processor **120**. For example, the processor **120** may change a width of the pulse signal **594** to indicate a start timing (or a timing) of an emission synchronization signal **592** for the processor **120**. For example, the processor **120** may change a width of the pulse signal **594** to indicate a start timing of a vertical synchronization signal **593** for the processor **120**. For example, the processor **120** may transmit a pulse signal **594** with a width **511** different from a width **512** and a width **513** to the display driver circuit **131** at a start timing **521** of the horizontal synchronization signal **591** that does not overlap with the start timing of the emission synchronization signal **592** and the start timing of the vertical synchronization signal **593**, transmit the pulse signal **594** with the width **512** different from the width **511** and the width **513** to the display driver circuit **131** at the start timing **522** of the horizontal synchronization signal **591** that overlaps with the start timing of the emission synchronization signal **592** and does not overlap with the start timing of the vertical synchronization signal **593**, and transmit the pulse signal **594** with the width **513** different from the width **511** and the width **512** at the start timing **523** of the horizontal synchronization signal **591** that overlaps with the start timing of the emission synchronization signal **592** and the start timing of the vertical synchronization signal **593**.
(43) For example, the processor **120** may change a timing for changing a waveform of the pulse signal in response to a change in a timing of a synchronization signal. For example, a timing (or a start timing) of the vertical synchronization signal **593** may be changed. For example, when obtaining an image to be displayed on the display panel **132** and/or rendering the image to be displayed on the display panel **132** is delayed, the timing of the vertical synchronization signal **593** may be changed. As a non-limiting example, the delay may be caused by a timing at which user input is received. As a non-limiting example, the delay may be caused by a load of the processor **120**. As a non-limiting example, the delay may be caused by the number of layers configuring an image to be displayed on the display panel **132**. For another example, the timing of the vertical synchronization signal **593** may be changed according to a change in a refresh rate. However, it is not limited thereto.
(44) For example, the processor **120** may change the start timing of the vertical synchronization signal **593** from a targeted timing **551** to a timing **552**. Each of the targeted timing **551** and the timing **552** may overlap the start timing of the emission synchronization signal **592** for the processor **120**. For example, the timing **552** may be identified from among the start timing of the emission synchronization signal **592** after the targeted timing **551**.
(45) For example, the processor **120** may change a waveform of the pulse signal **594** based on identifying the start timing of the vertical synchronization signal **593** as the timing **552** changed from the targeted timing **551**. For example, the processor **120** may change a timing of transmitting the pulse signal **594** with the width **513**, based on changing the start timing of the vertical

synchronization signal **593** to the timing **552** changed from the targeted timing **551**. For example, the processor **120** may refrain from or bypass transmitting the pulse signal **594** with the width **513** to the display driver circuit **131** at the targeted timing **551**, and transmit the pulse signal **594** having the width **513** to the display driver circuit **131** at the timing **552**, based on identifying the start timing of the vertical synchronization signal **593** as the timing **552** changed from the targeted timing **551**. For example, the processor **120** may request or inform the display driver circuit **131** to change the timing of the vertical synchronization signal for the display driver circuit **131**, by transmitting the pulse signal **594** with the width **513** to the display driver circuit **131** at the timing **552**.

(46) As a non-limiting example, after the start timing of the vertical synchronization signal **593** is changed to the timing **552**, the cycle **503** of the vertical synchronization signal **593** may be restored. For example, when the cycle **503** is restored, the processor **120** may transmit the pulse signal **594** with the width **513** for each the cycle **503**.

(47) Referring back to FIG. **1**, the pulse signal may be transmitted from the processor **120** to the display driver circuit **131**, to provide the display driver circuit **131** with a control command (or a command) related to a display on the display panel **132**. For example, the control command may be indicated by changing the waveform (or width) of the pulse signal. For example, the processor **120** may identify the control command to be provided to the display driver circuit **131** with respect to the display of the image on the display panel **132**. For example, the processor **120** may identify the control command to be synchronized with an image on the display panel **132**. For example, the processor **120** may change the waveform of the pulse signal periodically transmitted to the display driver circuit **131**, in order to indicate the control command to the display driver circuit **131** based on the identification. For example, the control command may include a control command (e.g., still indication, sticky flag indication, and/or on-the-fly indication) indicating that the image is stored in GRAM (e.g., the GRAM **220** of FIG. **2**) in the display driver circuit **131**.

(48) For example, the control command may include a control command indicating that a refresh rate (or a refresh rate of the image provided from the processor **120** to display driver circuit **131**) of the display panel **132** is changed. For example, the control command may include a control command indicating a change in an interface with respect to the image. However, it is not limited thereto.

(49) The waveform of the pulse signal changed to indicate the control command may be illustrated by way of non-limiting example below with reference to FIG. **6**.

(50) FIG. **6** illustrates an exemplary method of changing a waveform of a pulse signal to indicate a control command.

(51) Referring to FIG. **6**, the processor **120** may transmit a first image to the display driver circuit **131** through a first path **201**, as shown in a state **601**. The display driver circuit **131** may display the first image received from the processor **120** through the first path **201** on the display panel **132**, as indicated by an arrow **611**.

(52) The processor **120** may transmit a second image to the display driver circuit **131** through a first path **201**, as shown in a state **602**. The display driver circuit **131** may display the second image received from the processor **120** through the first path **201** on the display panel **132**, as indicated by an arrow **612**. As a non-limiting example, the second image may be at least partially different from the first image. As a non-limiting example, the second image may be the same as the first image. For example, the second image may be the first image transmitted again from the processor **120**.

(53) The processor **120** may identify to provide a control command indicating that the second image is stored in the GRAM **220** to the display driver circuit **131**, based on the second image.

(54) For example, the control command may be identified based on identifying that a time when the first image is displayed on the display panel **132** is longer than or equal to a reference time. However, it is not limited thereto. For example, the processor **120** may provide the control command to the display driver circuit **131** by changing a width of the pulse signal **692** transmitted

to the display driver circuit **131** based on a cycle **690** through the second path **202**. For example, the processor **120** may provide the control command to display driver circuit **131**, by changing a width of the pulse signal **692** transmitted at a timing **694** before the second image is provided to the display driver circuit **131** from a width **693** to a width **695**. For example, the timing **694** may be before a start timing **696** of the vertical synchronization signal **691** for the second image. For example, the display driver circuit **131** may receive the pulse signal **692** with the width **695** from the processor **120**. For example, the display driver circuit **131** may store the second image in the GRAM **220**, in response to the pulse signal **692** with the width **695** different from the width **693**, as indicated by an arrow **613**. For example, the display driver circuit **131** may scan the second image stored in the GRAM **220**, based on a change from the first image to the second image (and/or the pulse signal **692** with the width **695**), as indicated by an arrow **614**. For example, the display driver circuit **131** may display the second image again on the display panel **132**, based on scanning the second image, as indicated by an arrow **615**. For example, displaying the second image on the display panel **132** again may be executed in order to reduce an afterimage that may be caused on the display panel **132** according to the change from the first image to the second image. However, it is not limited thereto.

(55) Although not illustrated in FIG. **6**, the pulse signal **692** may further indicate the start timing (or a timing) of the vertical synchronization signal **691**. For example, the processor **120** may transmit the pulse signal **692** with a width different from the width **693** and the width **695**, to the display driver circuit **131** through the second path **202**, at the start timing of the horizontal synchronization signal for the processor **120** overlapping (or corresponding) to the start timing **696** of the vertical synchronization signal **691**.

(56) Although not illustrated in FIG. **6**, the pulse signal **692** may further indicate the start timing of the emission synchronization signal for the processor **120**. For example, the processor **120** may transmit the pulse signal **692** with a width to the display driver circuit **131** through the second path **202** at the start timing of the horizontal synchronization signal overlapping the start timing of the emission synchronization signal, wherein the width being different from the width **693**, the width **695**, and a width to indicate the start timing **696** of the vertical synchronization signal **691**.

(57) FIG. **7** is a block diagram illustrating an electronic device **701** in a network environment **700** according to various embodiments. Referring to FIG. **7**, the electronic device **701** in the network environment **700** may communicate with an electronic device **702** via a first network **798** (e.g., a short-range wireless communication network), or at least one of an electronic device **704** or a server **708** via a second network **799** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **701** may communicate with the electronic device **704** via the server **708**. According to an embodiment, the electronic device **701** may include a processor **720**, memory **730**, an input module **750**, a sound output module **755**, a display module **760**, an audio module **770**, a sensor module **776**, an interface **777**, a connecting terminal **778**, a haptic module **779**, a camera module **780**, a power management module **788**, a battery **789**, a communication module **790**, a subscriber identification module (SIM) **796**, or an antenna module **797**. In various embodiments, at least one of the components (e.g., the connecting terminal **778**) may be omitted from the electronic device **701**, or one or more other components may be added in the electronic device **701**. In various embodiments, some of the components (e.g., the sensor module **776**, the camera module **780**, or the antenna module **797**) may be implemented as a single component (e.g., the display module **760**).

(58) The processor **720** may execute, for example, software (e.g., a program **740**) to control at least one other component (e.g., a hardware or software component) of the electronic device **701** coupled with the processor **720**, and may perform various data processing or computation.

(59) According to an embodiment, as at least part of the data processing or computation, the processor **720** may store a command or data received from another component (e.g., the sensor module **776** or the communication module **790**) in volatile memory **732**, process the command or

the data stored in the volatile memory **732**, and store resulting data in non-volatile memory **734**. According to an embodiment, the processor **720** may include a main processor **721** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **723** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **721**. For example, when the electronic device **701** includes the main processor **721** and the auxiliary processor **723**, the auxiliary processor **723** may be adapted to consume less power than the main processor **721**, or to be specific to a specified function. The auxiliary processor **723** may be implemented as separate from, or as part of the main processor **721**.

(60) The auxiliary processor **723** may control at least some of functions or states related to at least one component (e.g., the display module **760**, the sensor module **776**, or the communication module **790**) among the components of the electronic device **701**, instead of the main processor **721** while the main processor **721** is in an inactive (e.g., sleep) state, or together with the main processor **721** while the main processor **721** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **723** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **780** or the communication module **790**) functionally related to the auxiliary processor **723**. According to an embodiment, the auxiliary processor **723** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **701** where the artificial intelligence is performed or via a separate server (e.g., the server **708**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

(61) The memory **730** may store various data used by at least one component (e.g., the processor **720** or the sensor module **776**) of the electronic device **701**. The various data may include, for example, software (e.g., the program **740**) and input data or output data for a command related thereto. The memory **730** may include the volatile memory **732** or the non-volatile memory **734**.

(62) The program **740** may be stored in the memory **730** as software, and may include, for example, an operating system (OS) **742**, middleware **744**, or an application **746**.

(63) The input module **750** may receive a command or data to be used by another component (e.g., the processor **720**) of the electronic device **701**, from the outside (e.g., a user) of the electronic device **701**. The input module **750** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

(64) The sound output module **755** may output sound signals to the outside of the electronic device **701**. The sound output module **755** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

(65) The display module **760** may visually provide information to the outside (e.g., a user) of the electronic device **701**. The display module **760** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **760** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force

incurred by the touch.
(66) The audio module **770** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **770** may obtain the sound via the input module **750**, or output the sound via the sound output module **755** or a headphone of an external electronic device (e.g., an electronic device **702**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **701**.
(67) The sensor module **776** may detect an operational state (e.g., power or temperature) of the electronic device **701** or an environmental state (e.g., a state of a user) external to the electronic device **701**, and then generate an electrical signal or data value corresponding to the detected state.
(68) According to an embodiment, the sensor module **776** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.
(69) The interface **777** may support one or more specified protocols to be used for the electronic device **701** to be coupled with the external electronic device (e.g., the electronic device **702**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **777** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.
(70) A connecting terminal **778** may include a connector via which the electronic device **701** may be physically connected with the external electronic device (e.g., the electronic device **702**). According to an embodiment, the connecting terminal **778** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).
(71) The haptic module **779** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **779** may include, for example, a motor, a piezoelectric element, or an electric stimulator.
(72) The camera module **780** may capture a still image or moving images. According to an embodiment, the camera module **780** may include one or more lenses, image sensors, image signal processors, or flashes.
(73) The power management module **788** may manage power supplied to the electronic device **701**. According to an embodiment, the power management module **788** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).
(74) The battery **789** may supply power to at least one component of the electronic device **701**. According to an embodiment, the battery **789** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.
(75) The communication module **790** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **701** and the external electronic device (e.g., the electronic device **702**, the electronic device **704**, or the server **708**) and performing communication via the established communication channel. The communication module **790** may include one or more communication processors that are operable independently from the processor **720** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **790** may include a wireless communication module **792** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **794** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **798** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **799** (e.g., a long-range communication network, such as a legacy cellular network, a 5G

network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **792** may identify and authenticate the electronic device **701** in a communication network, such as the first network **798** or the second network **799**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **796**.

(76) The wireless communication module **792** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **792** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **792** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **792** may support various requirements specified in the electronic device **701**, an external electronic device (e.g., the electronic device **704**), or a network system (e.g., the second network **799**). According to an embodiment, the wireless communication module **792** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 764 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 7 ms or less) for implementing URLLC.

(77) The antenna module **797** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **701**. According to an embodiment, the antenna module **797** may include an antenna including a radiating element including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **797** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **798** or the second network **799**, may be selected, for example, by the communication module **790** (e.g., the wireless communication module **792**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **790** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **797**.

(78) According to various embodiments, the antenna module **797** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

(79) At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

(80) According to an embodiment, commands or data may be transmitted or received between the electronic device **701** and the external electronic device **704** via the server **708** coupled with the second network **799**. Each of the electronic devices **702** or **704** may be a device of a same type as, or a different type, from the electronic device **701**. According to an embodiment, all or some of

operations to be executed at the electronic device **701** may be executed at one or more of the external electronic devices **702**, **704**, or **708**. For example, if the electronic device **701** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **701**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **701**. The electronic device **701** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **701** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In an embodiment, the external electronic device **704** may include an internet-of-things (IoT) device. The server **708** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **704** or the server **708** may be included in the second network **799**. The electronic device **701** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

(81) FIG. **8** is a block diagram **800** illustrating the display module **760** according to various embodiments. Referring to FIG. **8**, the display module **760** may include a display **810** and a display driver integrated circuit (DDI) **830** to control the display **810**. The DDI **830** may include an interface module **831**, memory **833** (e.g., buffer memory), an image processing module **835**, or a mapping module **837**. The various modules of the DDI may include various processing circuitry and/or executable program instructions. The DDI **830** may receive image information that contains image data or an image control signal corresponding to a command to control the image data from another component of the electronic device **701** via the interface module **831**. For example, according to an embodiment, the image information may be received from the processor **720** (e.g., the main processor **721** (e.g., an application processor)) or the auxiliary processor **723** (e.g., a graphics processing unit) operated independently from the function of the main processor **721**. The DDI **830** may communicate, for example, with touch circuitry **750** or the sensor module **776** via the interface module **831**. The DDI **830** may also store at least part of the received image information in the memory **833**, for example, on a frame by frame basis. The image processing module **835** may perform pre-processing or post-processing (e.g., adjustment of resolution, brightness, or size) with respect to at least part of the image data. According to an embodiment, the pre-processing or post-processing may be performed, for example, based at least in part on one or more characteristics of the image data or one or more characteristics of the display **810**. The mapping module **837** may generate a voltage value or a current value corresponding to the image data pre-processed or post-processed by the image processing module **835**. According to an embodiment, the generating of the voltage value or current value may be performed, for example, based at least in part on one or more attributes of the pixels (e.g., an array, such as an RGB stripe or a pentile structure, of the pixels, or the size of each subpixel). At least some pixels of the display **810** may be driven, for example, based at least in part on the voltage value or the current value such that visual information (e.g., a text, an image, or an icon) corresponding to the image data may be displayed via the display **810**.

(82) According to an embodiment, the display module **760** may further include the touch circuitry **850**. The touch circuitry **850** may include a touch sensor **851** and a touch sensor IC **853** to control the touch sensor **851**. The touch sensor IC **853** may control the touch sensor **851** to sense a touch input or a hovering input with respect to a certain position on the display **810**. To achieve this, for example, the touch sensor **851** may detect (e.g., measure) a change in a signal (e.g., a voltage, a quantity of light, a resistance, or a quantity of one or more electric charges) corresponding to the

certain position on the display **810**. The touch circuitry **850** may provide input information (e.g., a position, an area, a pressure, or a time) indicative of the touch input or the hovering input detected via the touch sensor **851** to the processor **720**. According to an embodiment, at least part (e.g., the touch sensor IC **853**) of the touch circuitry **850** may be formed as part of the display **810** or the DDI **830**, or as part of another component (e.g., the auxiliary processor **723**) disposed outside the display module **760**.
(83) According to an embodiment, the display module **760** may further include at least one sensor (e.g., a fingerprint sensor, an iris sensor, a pressure sensor, or an illuminance sensor) of the sensor module **776** or a control circuit for the at least one sensor. In such a case, the at least one sensor or the control circuit for the at least one sensor may be embedded in one portion of a component (e.g., the display **810**, the DDI **830**, or the touch circuitry **750**)) of the display module **760**. For example, when the sensor module **776** embedded in the display module **760** includes a biometric sensor (e.g., a fingerprint sensor), the biometric sensor may obtain biometric information (e.g., a fingerprint image) corresponding to a touch input received via a portion of the display **810**.
(84) As another example, when the sensor module **776** embedded in the display module **760** includes a pressure sensor, the pressure sensor may obtain pressure information corresponding to a touch input received via a partial or whole area of the display **810**. According to an embodiment, the touch sensor **851** or the sensor module **776** may be disposed between pixels in a pixel layer of the display **810**, or over or under the pixel layer.
(85) As described above, an electronic device may comprise: a processor, a display including a display driver circuit and a display panel, a first path connecting the display driver circuit to the processor, and a second path connecting the display driver circuit to the processor and separated from the first path. According to an example embodiment, the processor **120** may be configured to transmit, based on a first cycle, via the second path to the display driver circuit, a pulse signal to synchronize, with a first time period in the processor used to display an image transmitted via the first path to the display driver circuit on the display panel, the first time period in the display driver circuit used to display the image on the display panel. According to an example embodiment, the processor may be configured to change, based on a second cycle different from the first cycle, a waveform of the pulse signal transmitted from the first cycle from a first waveform to a second waveform to synchronize, with a second time period in the processor used for the display of the image, the second time period in the display driver circuit used for the display of the image.
(86) According to an example embodiment, the first path may be disabled in at least portion of a time period that ceases to transmit an image from the processor to the display driver circuit. According to an example embodiment, the processor may be configured to maintain, while the first path is disabled, transmitting the pulse signal via the second path to the display driver circuit.
(87) According to an example embodiment, the display driver circuit may include a graphic random access memory (GRAM). According to an example embodiment, the first path may be disabled in at least portion of a time period that executes a display on the display panel by scanning, by the display driver circuit, an image in the GRAM, wherein the image received from the processor. According to an example embodiment, the processor may be configured to maintain, while the first path is disabled, transmitting the pulse signal via the second path to the display driver circuit.
(88) According to an example embodiment, the second time period may be longer than the first time period.
(89) According to an example embodiment, the first time period may correspond to a cycle of a horizontal synchronization signal. According to an example embodiment, the second time period may correspond to a cycle of a vertical synchronization signal. According to an example embodiment, the processor may be configured to transmit, via the second path to the display driver circuit, the pulse signal with the second waveform, at first transmission timings each corresponding to a start timing of the vertical synchronization signal used in the processor from among transmission timings according to the first cycle. According to an example embodiment, the

processor may be configured to transmit, via the second path to the display driver circuit, the pulse signal with the first waveform, at second transmission timings each different from the start timing from among the transmission timings.
(90) According to an example embodiment, the first time period may correspond to a cycle of a horizontal synchronization signal. According to an example embodiment, the second time period may correspond to a cycle of emission synchronization signal indicating a transmission timing of an emission signal from the display driver circuit to the display panel. According to an example embodiment, the processor may be configured to transmit, via the second path to the display driver circuit, the pulse signal with the second waveform, at first transmission timings each corresponding to a start timing of the emission synchronization signal used in the processor from among transmission timings according to the first cycle. According to an example embodiment, the processor may be configured to transmit, via the second path to the display driver circuit, the pulse signal with the first waveform, at second transmission timings each different from the start timing from among the transmission timings.
(91) According to an example embodiment, the processor may be configured to change the waveform from the first waveform to the second waveform by changing a width of the pulse signal.
(92) According to an example embodiment, the processor may be configured to identify a control command to be provided to the display driver circuit with respect to the display of the image on the display panel. According to an example embodiment, the processor may be configured to change the waveform to a third waveform different from the first waveform and the second waveform to indicate the control command to the display driver circuit.
(93) According to an example embodiment, the control command may comprise a control command indicating to store the image in the GRAM.
(94) According to an example embodiment, the control command may comprise a control command indicating to change a refresh rate of the image provided from the processor to the display driver circuit.
(95) As described above, according to an example embodiment, an electronic device **100** may comprise: a processor, a display including a display driver circuit and a display panel, a first path connecting the display driver circuit to the processor, a second path connecting the display driver circuit to the processor and separated from the first path. According to an example embodiment, the processor may be configured to periodically transmit, via the second path to the display driver circuit, a pulse signal to synchronize, with a time period in the processor used to display on the display panel an image transmitted via the first path to the display driver circuit, the time period in the display driver circuit used to display on the display panel the image. According to an example embodiment, the processor may be configured to identify a control command to be provided to the display driver circuit with respect to the display of the image on the display panel. According to an example embodiment, the processor may be configured to, based on the identification, change a waveform of the pulse signal that is periodically transmitted to the display driver circuit from a first waveform to a second waveform to indicate the control command to the display driver circuit.
(96) According to an example embodiment, the display driver circuit may include a graphic random access memory (GRAM). According to an example embodiment, the control command may comprise a control command indicating to store the image in the GRAM.
(97) According to an example embodiment, the control command may comprise a control command indicating to change a refresh rate of the image provided from the processor to the display driver circuit.
(98) According to an example embodiment, the first path may be disabled in at least portion of a time period that ceases to transmit an image from the processor to the display driver circuit. According to an example embodiment, the processor may be configured to maintain to transmit the pulse signal via the second path to the display driver circuit, while the first path is disabled.
(99) According to an example embodiment, the first path may be disabled in at least portion of a

time period that executes a display on the display panel by scanning, by the display driver circuit, an image in the GRAM, wherein the image received from the processor. According to an example embodiment, the processor may be configured to maintain transmitting the pulse signal via the second path to the display driver circuit, while the first path is disabled.
(100) According to an example embodiment, the processor may be configured to periodically transmit the pulse signal by transmitting the pulse signal based on the first cycle. According to an example embodiment, the processor may be configured to change, based on a second cycle different from the first cycle, a waveform of the pulse signal transmitted based on the first cycle from a first waveform or a second waveform to a third waveform to synchronize, with another time period in the processor used for the display of the image, the other time period in the display driver circuit used for the display of the image.
(101) According to an example embodiment, the other time period may be longer than the time period.
(102) According to an example embodiment, the time period may correspond to a cycle of a horizontal synchronization signal. According to an example embodiment, the other time period may correspond to a vertical synchronization signal. According to an example embodiment, the processor may be configured to transmit, via the second path to the display driver circuit, the pulse signal with the third waveform, at first transmission timings each corresponding to a start timing of the vertical synchronization signal used in the processor from among transmission timings according to the first cycle. According to an example embodiment, the processor may be configured to transmit, via the second path to the display driver circuit, the pulse signal with the first waveform, at second transmission timings each different from the start timing from among the transmission timings.
(103) According to an example embodiment, the second transmission timings may be different from third transmission timings for indicating the control command from among the transmission timings.
(104) According to an example embodiment, the processor **120** may be configured to change the waveform from the first waveform to the second waveform by changing a width of the pulse signal.
(105) The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, a home appliance, or the like. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.
(106) It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.
(107) As used in connection with various embodiments of the disclosure, the term “module” may

include a unit implemented in hardware, software, or firmware, or any combination thereof, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

(108) Various embodiments as set forth herein may be implemented as software (e.g., the program **740**) including one or more instructions that are stored in a storage medium (e.g., internal memory **736** or external memory **738**) that is readable by a machine (e.g., the electronic device **701**). For example, a processor (e.g., the processor **720**) of the machine (e.g., the electronic device **701**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the “non-transitory” storage medium is a tangible device, and may not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

(109) According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server. According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

Claims

1. An electronic device comprising: at least one processor comprising processing circuitry; memory comprising one or more storage media storing instructions; a display including display driver circuitry and a display panel; a first path, connecting the display driver circuitry to the at least one processor, configured to transmit an image from the at least one processor to the display driver circuitry; and a second path, connecting the display driver circuitry to the at least one processor, separated from the first path, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: periodically transmit, via the second path, from the at least one processor to the display driver circuitry, a pulse signal to synchronize

counting of the at least one processor performed for displaying an image on the display panel with counting of the display driver circuitry performed for displaying an image on the display panel; in response to a start timing of a first synchronization signal for the at least one processor being identified as not overlapping a start timing of a second synchronization signal for the at least one processor in accordance with the counting of the at least one processor, set a waveform of the pulse signal periodically transmitted as a first waveform; and in response to a start timing of the first synchronization signal for the at least one processor being identified as overlapping a start timing of the second synchronization signal for the at least one processor in accordance with the counting of the at least one processor, set the waveform of the pulse signal periodically transmitted as a second waveform different from the first waveform, and wherein a cycle of the first synchronization signal for the at least one processor is shorter than a cycle of the second synchronization signal for the at least one processor.

2. The electronic device of claim 1, wherein the first path is configured to be disabled in at least portion of a time period that ceases to transmit an image from the at least one processor to the display driver circuitry, and wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to maintain, while the first path is disabled, the pulse signal periodically transmitted via the second path from the at least one processor to the display driver circuitry.

3. The electronic device of claim 1, wherein the display driver circuitry includes a graphic random access memory (GRAM), wherein the first path is configured to be disabled in at least portion of a time period that executes displaying on the display panel by scanning, by the display driver circuitry, an image stored in the GRAM, and wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to maintain, while the first path is disabled, the pulse signal periodically transmitted via the second path from the at least one processor to the display driver circuitry.

4. The electronic device of claim 1, wherein the display driver circuitry is configured to: based on the waveform of the pulse signal periodically transmitted from the at least one processor to the display driver circuitry being identified as the first waveform, synchronize the first synchronization signal for the display driver circuitry with the first synchronization signal for the at least one processor; and based on the waveform of the pulse signal periodically transmitted from the at least one processor to the display driver circuitry being identified as the second waveform, synchronize the second synchronization signal for the display driver circuitry with the second synchronization signal for the at least one processor.

5. The electronic device of claim 1, wherein the first synchronization signal for the at least one processor comprises a horizontal synchronization signal for the at least one processor, and wherein the second synchronization signal for the at least one processor comprises a vertical synchronization signal for the at least one processor.

6. The electronic device of claim 1, wherein the first synchronization signal for the at least one processor comprises a horizontal synchronization signal for the at least one processor, and wherein the second synchronization signal for the at least one processor comprises an emission synchronization signal for the at least one processor indicating a transmission timing of an emission signal from the display driver circuitry to the display panel.

7. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: set the waveform of the pulse signal periodically transmitted as the first waveform by setting a width of the pulse signal as a first width; and set the waveform of the pulse signal periodically transmitted as the second waveform by setting the width of the pulse signal as a second width different from the first width.

8. The electronic device of claim 1, wherein a transmission cycle of the pulse signal periodically transmitted corresponds to the cycle of the first synchronization signal for the at least one processor.

9. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: identify a control command to be provided to the display driver circuitry with respect to displaying an image on the display panel; and based on the identification, provide the control command to the display driver circuitry by setting the waveform of the pulse signal periodically transmitted as a third waveform different from the first waveform and the second waveform.
10. The electronic device of claim 9, wherein the display driver circuit includes a graphic random access memory (GRAM), and wherein the control command comprises a control command indicating to store an image in the GRAM and/or a control command indicating to change a refresh rate of an image to be displayed on the display panel.
11. An electronic device comprising: at least one processor comprising processing circuitry; a display including display driver circuitry and a display panel; a first path, connecting the display driver circuitry to the at least one processor, configured to transmit an image from the at least one processor to the display driver circuitry; and a second path, connecting the display driver circuitry to the at least one processor, separated from the first path, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: periodically transmit, via the second path, from the at least one processor to the display driver circuitry, a pulse signal to synchronize counting of the at least one processor performed for displaying an image on the display panel; with counting of the display driver circuitry performed for displaying an image on the display panel; identify a control command to be provided to the display driver circuit with respect to the display of the image on the display panel; based on the identification, provide the control command to the display driver circuitry by changing a waveform of the pulse signal periodically transmitted from a first waveform to a second waveform different from the first waveform; based on providing the control command to the display driver circuitry, restore the waveform of the pulse signal periodically transmitted to the first waveform.
12. The electronic device of claim 11, wherein the display driver circuit includes a graphic random access memory (GRAM), and wherein the control command comprises a control command indicating storing an image in the GRAM.
13. The electronic device of claim 11, wherein the control command comprises a control command indicating changing a refresh rate of an image to be displayed on the display panel.
14. The electronic device of claim 11, wherein the first path is configured to be disabled in at least portion of a time period that ceases to transmit an image from the at least one processor to the display driver circuitry, and wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to maintain the pulse signal periodically transmitted via the second path from the at least one processor to the display driver circuitry, while the first path is disabled.
15. The electronic device of claim 11, wherein the display driver circuit includes a graphic random access memory (GRAM), wherein the first path is configured to be disabled in at least portion of a time period that executes displaying on the display panel by scanning, by the display driver circuitry, an image stored in the GRAM, and wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to maintain, while the first path is disabled, the pulse signal periodically transmitted via the second path from the at least one processor to the display driver circuitry.
16. The electronic device of claim 11, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: in response to a start timing of a first synchronization signal for the at least one processor being identified as not overlapping a start timing of a second synchronization signal for the at least one processor in accordance with the counting of the at least one processor, set the waveform of the pulse signal periodically transmitted as the first waveform; and in response to a start timing of the first synchronization signal for the at least one processor being identified as overlapping a start timing of the second synchronization

signal for the at least one processor in accordance with the counting of the at least one processor, set the waveform of the pulse signal periodically transmitted as a third waveform different from the first waveform and the second waveform, and wherein a cycle of the first synchronization signal for the at least one processor is shorter than a cycle of the second synchronization signal for the at least one processor.

17. The electronic device of claim 16, wherein the display driver circuitry is configured to: based on the waveform of the pulse signal periodically transmitted from the at least one processor to the display driver circuitry being identified as the first waveform, synchronize the first synchronization signal for the display driver circuitry with the first synchronization signal for the at least one processor; and based on the waveform of the pulse signal periodically transmitted from the at least one processor to the display driver circuitry being identified as the third waveform, synchronize the second synchronization signal for the display driver circuitry with the second synchronization signal for the at least one processor.

18. The electronic device of claim 17, wherein the first synchronization signal for the at least one processor comprises a horizontal synchronization signal for the at least one processor, and wherein the second synchronization signal for the at least one processor comprises a vertical synchronization signal for the at least one processor.

19. The electronic device of claim 18, wherein the control command is provided before a start timing of the vertical synchronization signal for the at least one processor used for displaying an image to which the control command is to be applied.

20. An electronic device comprising: at least one processor comprising processing circuitry; memory comprising one or more storage media storing instructions; a display including display driver circuitry and a display panel; a first path, connecting the display driver circuitry to the at least one processor, configured to transmit an image from the at least one processor to the display driver circuitry; and a second path, connecting the display driver circuitry to the at least one processor, separated from the first path, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: periodically transmit, via the second path, from the at least one processor to the display driver circuitry, a pulse signal to synchronize counting of the at least one processor performed for displaying an image on the display panel with counting of the display driver circuitry performed for displaying an image on the display panel, wherein a transmission cycle of the pulse signal periodically transmitted corresponds to a cycle of a first synchronization signal for the at least one processor; based on a start timing of the first synchronization signal for the at least one processor being identified as not overlapping a start timing of a second synchronization signal for the at least one processor in accordance with the counting of the at least one processor, maintain a waveform of the pulse signal periodically transmitted as a first waveform; and based on a start timing of the first synchronization signal for the at least one processor being identified as overlapping a start timing of the second synchronization signal for the at least one processor in accordance with the counting of the at least one processor, change the waveform of the pulse signal periodically transmitted to a second waveform different from the first waveform.